Can Data Beat the Market? Predicting NBA Games and Evaluating Kalshi Odds

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Summary of Research Questions

1. **How accurately do Kalshi's pregame NBA prices represent the actual probability of each team winning?**
   I will compare Kalshi's market-implied probabilities shortly before each game begins with the actual game results. I will determine whether contracts priced near 60%, for example, win approximately 60% of the time.
2. **Can an NBA prediction model outperform Kalshi's market probabilities?**
   I will create machine-learning models using information available before each game, including recent win percentage, average point differential, home-court advantage, rest days, and team rating. I will compare each model's accuracy, Brier score, and log loss with Kalshi's pregame probabilities.
3. **Under what conditions are Kalshi's NBA predictions most and least accurate?**
   I will examine whether prediction accuracy changes based on market volume, probability range, home-court status, rest differences, and differences in recent team performance. This will help determine whether certain types of games are more difficult for the market to price.
4. **Would differences between the model and Kalshi's prices produce positive simulated returns?**
   When the model's probability differs from Kalshi's price by at least 5%, 10%, or 15%, I will simulate purchasing one contract. I will calculate the resulting profit, loss, and return on investment without placing any real trades.

Motivation

Prediction markets such as Kalshi allow participants to trade contracts based on whether specific events will occur. In an NBA game market, the price of a contract can be interpreted as the market's approximate probability that a team will win. However, these prices are influenced by traders, available information, market liquidity, and public opinion.

I am interested in whether statistical models can identify patterns that the market does not fully account for. NBA games are especially useful for this analysis because teams play frequently and provide a large amount of historical performance data. Factors such as recent form, home-court advantage, rest, and point differential can be measured before each game.

This project will function as a data-driven deep dive into Kalshi's NBA markets. It will not assume that a model can reliably generate profits. Instead, it will evaluate whether market prices are calibrated, when the market tends to make larger errors, and whether apparent predictive advantages remain when tested on games the model has never seen.

Dataset

Kalshi NBA Market Data

The first dataset will come from Kalshi's public API. Kalshi provides public market-data endpoints that can be accessed without authentication. The NBA game-winner markets belong to the **KXNBAGAME** series. Kalshi's official API documentation explains how to retrieve public market information.

I will retrieve recent markets from:

https://external-api.kalshi.com/trade-api/v2/markets?series_ticker=KXNBAGAME&limit=1000

Older markets will be retrieved from:

https://external-api.kalshi.com/trade-api/v2/historical/markets?series_ticker=KXNBAGAME&limit=1000

Kalshi separates recent and archived market records, so I will use its cutoff endpoint to determine which source contains each market. This process follows Kalshi's historical-data instructions.

Important fields include **ticker, event_ticker, yes_sub_title, no_sub_title, occurrence_datetime, settlement_value_dollars,** and **volume_fp.** I will also use Kalshi's historical candlestick endpoint to retrieve one-minute price histories. For each game, I will select the final available price at least 15 minutes before the scheduled starting time so that live-game information cannot enter the analysis.

Because the API uses nested responses and pagination, my Python program will repeatedly request pages until the returned cursor is empty. I will save the original API responses and include a shareable copy of the resulting data with the final submission.

NBA Schedule and Results Data

Game results will come from Basketball Reference's 2025-26 NBA schedule:

https://www.basketball-reference.com/leagues/NBA_2026_games.html

The schedule provides the date, visiting team, home team, visiting-team points, home-team points, overtime status, and links to monthly schedule pages. I will programmatically collect and combine the monthly tables rather than manually copying the information.

The Basketball Reference data will be matched with the Kalshi data using the game date, home team, and visiting team. I will create a team-name dictionary because the two sources may use different abbreviations or naming formats. All cleaning and merging will be completed through Python.

Challenge Goals

Messy Data: This project will collect data from a paginated API and HTML tables instead of relying on one prepared CSV file. The Kalshi information is divided between live and historical endpoints, while detailed price history requires a separate request for each market. I will write Python functions to request the data, process nested fields, follow pagination cursors, convert timestamps, normalize team names, and combine market records with NBA results.

Machine Learning: I will train and compare multiple classification models instead of relying on one basic model. The planned models include logistic regression, random forest, and gradient boosting. I will test multiple combinations of parameters, such as the number of estimators, maximum depth, learning rate, and regularization strength. The best model will be selected using validation data and evaluated once on a later chronological test set.

Method

Research Question 1: Kalshi Calibration

For each market, I will calculate the midpoint between the final pregame **yes_bid** and **yes_ask**. This midpoint will be treated as Kalshi's estimated probability for the team represented by the Yes contract.

I will use **settlement_value_dollars** and the Basketball Reference scores to determine the winner. The probabilities will be divided into ranges such as 0.40–0.49, 0.50–0.59, and 0.60–0.69. Within each range, I will compare the average market probability with the percentage of contracts that settled at $1. A calibration plot, Brier score, log loss, and prediction accuracy will show whether Kalshi's probabilities correspond closely with observed results.

Research Question 2: Model Performance

Games will be sorted chronologically. The earliest 60% will form the training set, the next 20% will form the validation set, and the latest 20% will be reserved as the test set. A chronological split will prevent future games from influencing predictions about earlier games.

For every team and game, I will calculate pregame features using only previously completed games:

- Win percentage over the previous 5 and 10 games
- Average point differential over the previous 5 and 10 games
- Current Elo-style team rating
- Number of rest days
- Whether the team is playing on consecutive days
- Whether the team is playing at home

The final model columns will represent the difference between the home and visiting teams for each feature. The target variable will equal 1 when the home team wins and 0 when the visiting team wins.

I will tune the models using the validation period and then compare their accuracy, Brier score, and log loss on the untouched test period. I will calculate the same metrics for Kalshi using exactly the same test games. This will definitively show whether any model performs better than the market.

Research Question 3: Conditions Affecting Accuracy

I will divide games into groups based on Kalshi volume, market probability, rest difference, home-court status, and differences in recent point differential. For each group, I will calculate the market's prediction accuracy and Brier score.

For example, I will compare the lowest-volume quartile with the highest-volume quartile. If high-volume markets have lower Brier scores, that would suggest that greater participation is associated with more accurate prices. These comparisons will be displayed with bar charts and scatterplots.

Research Question 4: Simulated Returns

For each game in the test period, I will calculate:

model probability - Kalshi purchase price

I will simulate buying a contract only when this estimated advantage passes a selected threshold. I will test thresholds of 5%, 10%, and 15%. Profit will be calculated using the contract's settlement value minus its purchase price and applicable transaction costs.

I will report the number of simulated positions, total profit or loss, return on investment, and cumulative results over time. This will be a historical simulation only and will not involve real-money trading.

Work Plan

1. **Download and inspect the datasets: 3 hours**
   Test the Kalshi endpoints, retrieve Basketball Reference tables, and confirm that both sources contain enough matching games.
2. **Clean and merge the data: 5 hours**
   Process API pagination, convert timestamps, standardize team names, remove incomplete records, and merge games across sources.
3. **Create pregame features: 5 hours**
   Calculate rolling performance, rest, home-court, point-differential, and Elo-rating variables without using future information.
4. **Train and evaluate models: 6 hours**
   Create chronological splits, tune multiple models, calculate evaluation metrics, and compare model performance with Kalshi.
5. **Create visualizations and write the report: 5 hours**
   Produce calibration plots, feature comparisons, model-performance charts, and simulated-return graphs.

I will complete the project individually using Python and VS Code. I will organize the analysis into separate modules for data collection, cleaning, feature construction, modeling, and visualization. I will use Git commits to preserve working versions and write tests for important functions, including team-name conversion, probability calculation, chronological splitting, and rolling-feature construction. If API access or historical coverage becomes a problem, I will save the raw data locally and limit the analysis to the date range shared by both sources.